

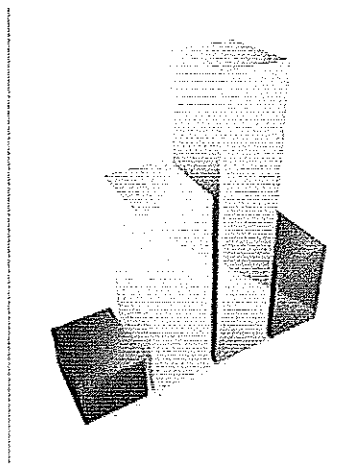
Full Name: SOLUTIONS



**MATHEMATICS
YEAR 9**

Investigation 1 – Range and Measures of Centre

Semester 1 2015



Answer Sheet

Information for Students

Working time:
Marks available:

50 minutes
30 marks

28??

Range and measures of centre

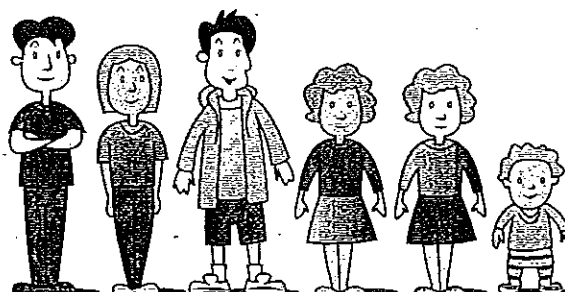


It is sometimes useful to summarise a large group of data as a single value. The concept of 'average' is familiar to most people, but more precise mathematical terms to use are 'mean', 'median' and 'mode'.

► Let's start: Family heights

Each New Year, the Green family measure and record their heights. This year their height measurements are:

Georgia 78 cm, Emily 130 cm, Amy 130 cm, Ethan 188 cm, Mrs Green 165 cm, Mr Green 182 cm.



Displaying Data

- 1 Draw a bar graph to show the heights of the people in the Green family.

Range

- 1 Who is the shortest and who is the tallest person in the Green family?
- 2 What is the range (the difference) between the shortest and tallest heights in the Green family?
- 3 The shortest person in the world is 59.93 cm and the tallest person in the world is 251 cm. What is the current range of heights for all adult humans?
- 4 The Green family had a snow-skiing holiday. One morning it is -8°C and that afternoon it is 5°C . What was the range of temperature that day?

Median

- 1 List the heights of the Green family in ascending (increasing) order.
- 2 What are the two middle heights?
- 3 Find the median (middle of these two central heights).
- 4 If the tallest man in the world, height 251 cm, is added into the Green family heights list, what is the median (middle) height now?
- 5 By how much has the median height changed by adding the tallest man into the list?
- 6 Does the median value always have to be one of the scores in the list?

Mean

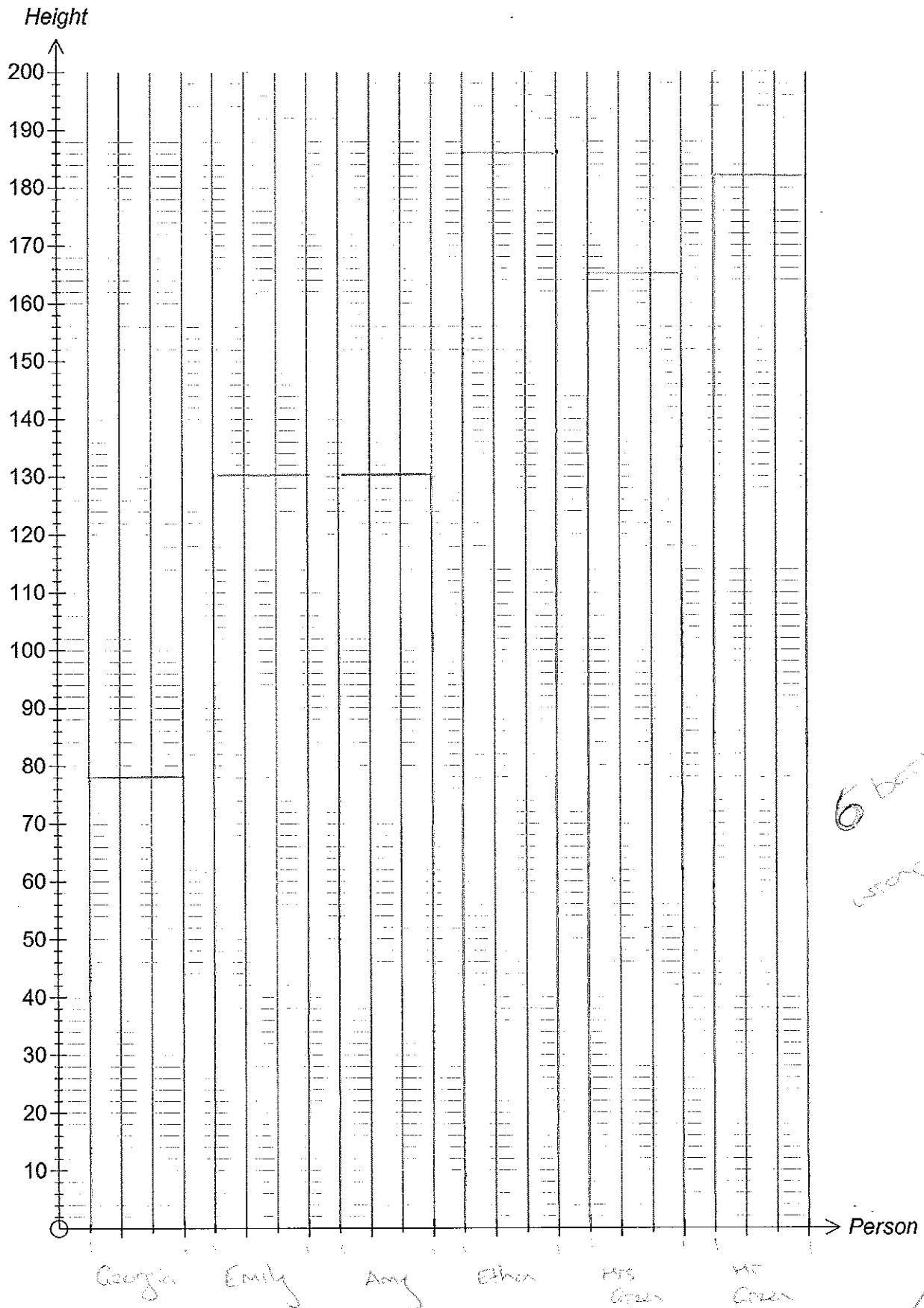
- 1 Add up the total of all the heights of the Green family.
- 2 Now find the mean height. (mean = total of heights divided by the number of heights)
- 3 If the tallest man in the world is included with the Green family heights, what is the mean height now?
- 4 By how much has the mean height changed by including the tallest man into the list?
- 5 Does the mean value always have to be one of the scores in the list?

Mode

- 1 The Green family has twins. Who are they and what is their height?
- 2 What is the mode (most common) of the Green family heights?
- 3 The Pink family has heights: 125 cm, 142 cm, 142 cm, 142 cm, 160 cm and 178 cm. What is the height of the Pink triplets?
- 4 What is the mode of the Pink family heights?
- 5 Does the mode value always have to be one of the scores in the list?

Write your answers in the spaces provided.

Displaying Data



6 bars = 6 marks
wrong graph (-2)

(1 mark for scale/number)

Range

1. George ✓ is shortest and Ethan ✓ is tallest.
2. 110 cm ✓
3. 191070 ✓ $251 - 57.93$
4. 13°C ✓

Median

1. 78 130 130 165 182 188 ✓
2. 130 165 ✓
3. 147.5 ✓
4. 165 ✓ $78 \ 130 \ 130 \ 165 \ 182 \ 188 \ 251$
5. up 17.5 ✓
6. n.b. it could be between values on the list ✓ Reasoning

Mean

1. 873 ✓
2. 145.5 cm ✓ $873 \div 6$
3. 160.6 cm ✓ $1124 \div 7$
4. 15.1 cm ✓
5. n.b. most of the time it will not be a value on the list ✓ Reasoning

Mode

1. Emily 130 cm ; Amy 130 cm. ✓
2. 130 cm ✓
3. 142 cm ✓
4. 142 cm ✓
5. yes ✓

End of Investigation

This space is for additional working.